

ASSIGNMENT-14

Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS

#TASK-1

AI-Assisted Portfolio Website

Scenario:

A student wants to showcase their projects, skills, and contact details in a portfolio website. Instead of writing all code manually, they want to speed up the process using GitHub Copilot.

- Use Copilot to generate an HTML structure for a personal portfolio page (sections: Home, About, Projects, Contact).
- Ask Copilot to suggest responsive CSS styling for the layout (e.g., grid/flexbox).
- Customize Copilot's suggestions to add a hover effect on project cards.

#PROMPT:

"Generate Python code that automatically creates a personal portfolio website. The Python program should:

- Create an index.html file with sections: Home, About, Projects, and Contact.
 - Create a style.css file with responsive design using Flexbox or Grid layout.
 - Add a hover effect on project cards.
 - Use clean, modern, and responsive HTML/CSS structure.
- The code should save both files in the current directory and print a confirmation message when done."

#CODE:

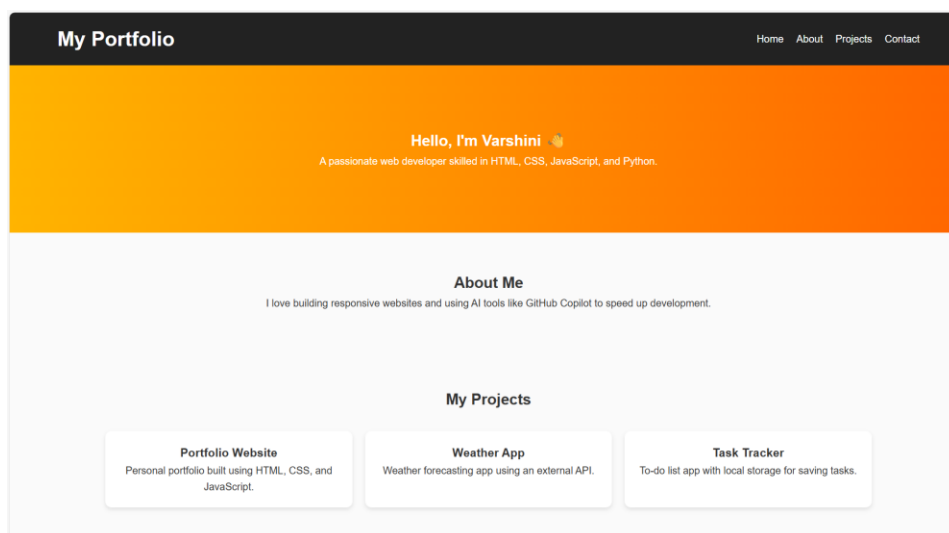
```
generated_sites > AI ASS-14.py > ...
1 # portfolio_generator.py
2
3 # Step 1: Define HTML Content
4 html_content = """
5 <!DOCTYPE html>
6 <html lang="en">
7 <head>
8   <meta charset="UTF-8">
9   <meta name="viewport" content="width=device-width, initial-scale=1.0">
10  <title>My Portfolio</title>
11  <link rel="stylesheet" href="style.css">
12 </head>
13 <body>
14   <header>
15     <nav>
16       <h1 class="logo">My Portfolio</h1>
17       <ul class="nav-links">
18         <li><a href="#home">Home</a></li>
19         <li><a href="#about">About</a></li>
20         <li><a href="#projects">Projects</a></li>
21         <li><a href="#contact">Contact</a></li>
22       </ul>
23     </nav>
24   </header>
25
26   <section id="home" class="section home">
27     <h2>Hello, I'm Varshini 🌟</h2>
28     <p>A passionate web developer skilled in HTML, CSS, JavaScript, and Python.</p>
29   </section>
30
31   <section id="about" class="section about">
32     <h2>About Me</h2>
33     <p>I love building responsive websites and using AI tools like GitHub Copilot to speed up development.</p>
34   </section>
35
36   <section id="projects" class="section projects">
37     <h2>My Projects</h2>
38     <div class="project-grid">
39       <div class="project-card">
40         <h3>Portfolio Website</h3>
41         <p>Personal portfolio built using HTML, CSS, and JavaScript.</p>
42       </div>
```

```

43     <div class="project-card">
44       <h3>Weather App</h3>
45       <p>Weather forecasting app using an external API.</p>
46     </div>
47     <div class="project-card">
48       <h3>Task Tracker</h3>
49       <p>To-do list app with local storage for saving tasks.</p>
50     </div>
51   </div>
52 </section>
53
54   <section id="contact" class="section contact">
55     <h2>Contact Me</h2>
56     <form>
57       <input type="text" placeholder="Your Name" required>
58       <input type="email" placeholder="Your Email" required>
59       <textarea placeholder="Your Message" required></textarea>
60       <button type="submit">Send</button>
61     </form>
62   </section>
63
64   <footer>
65     <p>© 2025 Varshini | All rights reserved.</p>
66   </footer>
67 </body>
68 </html>
69 """
70
71 # Step 2: Define CSS Content
```

```
generated_sites > AI ASS-14.py > ...
170  /* Footer */
171  footer {
172      background: #222;
173      color: white;
174      text-align: center;
175      padding: 10px 0;
176      margin-top: 20px;
177  }
178
179  /* Responsive */
180  @media (max-width: 600px) {
181      nav {
182          flex-direction: column;
183      }
184      .nav-links {
185          flex-direction: column;
186      }
187      .nav-links li {
188          margin: 10px 0;
189      }
190  }
191  ****
192
193  # Step 3: Write Files
194  with open("index.html", "w") as html_file:
195      |   html_file.write(html_content)
196
197  with open("style.css", "w") as css_file:
198      |   css_file.write(css_content)
199
200  print("✅ Portfolio website files created successfully: index.html and style.css")
201
```

#OUTPUT:



Files Used:

- index.html
- style.css
- script.js (optional)

TASK#2:

AI-Generated Restaurant Landing Page

Scenario:

A local restaurant needs a simple landing page with a navigation bar, menu highlights, and an image gallery. The developer wants to quickly generate it using AI assistance.

- Use Copilot to create a navigation bar with links (Home, Menu, Gallery, Contact).
- Generate a menu section styled with CSS cards.
- Add a JavaScript-based image slider for the gallery, with Copilot suggesting the base code.

#PROMPT

Generate a Python program that automatically creates a restaurant landing page.

The program should:

Create an index.html file with sections: **Home**, **Menu**, **Gallery**, and **Contact**.

Add a **navigation bar** with links to each section.

Design the **menu section** using **responsive CSS cards** for food items.

Add a **JavaScript image slider** for the gallery section that shows multiple food images with next/previous buttons.

Use clean, responsive **HTML**, **CSS**, and **JavaScript** code.

Save the generated files (index.html, style.css, script.js) in the current directory and print a success message after creation

#CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Restaurant Landing Page</title>
  <style>
    /* Basic styling */
    body {
      margin: 0;
      font-family: Arial, sans-serif;
    }

    /* Navigation bar */
    .navbar {
      background-color: #333;
      padding: 1rem;
    }

    .navbar a {
      color: white;
      text-decoration: none;
      padding: 1rem;
      margin: 0 10px;
    }

    /* Menu cards */
    .menu-section {
      display: flex;
      justify-content: center;
      flex-wrap: wrap;
      padding: 2rem;
    }

    .menu-card {
      width: 250px;
      margin: 1rem;
      padding: 1rem;
      border: 1px solid #ddd;
      border-radius: 8px;
      box-shadow: 0 2px 4px rgba(0,0,0,0.1);
    }
  </style>
</head>
<body>
```

```

    </div>
    <div class="menu-card">
      <h3>Grilled Salmon</h3>
      <p>Fresh salmon with seasonal vegetables</p>
      <p>$24.99</p>
    </div>
  </section>

  <!-- Image Slider -->
  <div class="slider-container">
    <div class="slide active">
      
    </div>
    <div class="slide">
      
    </div>
    <div class="slide">
      
    </div>
  </div>

  <script>
    // Image slider functionality
    let currentSlide = 0;
    const slides = document.querySelectorAll('.slide');

    function showSlide(n) {
      slides.forEach(slide => slide.classList.remove('active'));
      currentSlide = (n + slides.length) % slides.length;
      slides[currentSlide].classList.add('active');
    }

    // Auto-advance slides every 3 seconds
    setInterval(() => {
      showSlide(currentSlide + 1);
    }, 3000);
  </script>
</body>
```

#OUTPUT:

Specialty Pizza Fresh homemade pizza with premium toppings \$18.99	Pasta Alfredo Creamy pasta with garlic and parmesan \$15.99	Grilled Salmon Fresh salmon with seasonal vegetables \$24.99
---	--	---



Specialty Pizza Fresh homemade pizza with premium toppings \$18.99	Pasta Alfredo Creamy pasta with garlic and parmesan \$15.99	Grilled Salmon Fresh salmon with seasonal vegetables \$24.99
---	--	---



Specialty Pizza Fresh homemade pizza with premium toppings \$18.99	Pasta Alfredo Creamy pasta with garlic and parmesan \$15.99	Grilled Salmon Fresh salmon with seasonal vegetables \$24.99
---	--	---



#TASK-3:

AI-Powered Event Registration Form

Scenario:

SR University is hosting a tech fest. They need a web-based registration form for students. The form must validate user input in real-time.

- Ask Copilot to generate an HTML form (fields: Name, Email, Phone, Department, Event Selection).
- Use Copilot to assist in adding CSS styling for an attractive form layout.
- Implement JavaScript validation (e.g., email format check, phone number length check) using Copilot's suggestions.

#PROMPT:

- Create an index.html file with a registration form containing the following fields: **Name, Email, Phone, Department, and Event Selection (dropdown)**.
- Create a style.css file with modern, responsive styling for the form (centered layout, input box styling, and hover/submit button effects).
- Create a script.js file with **JavaScript-based real-time validation** for:
 - Checking **email format**
 - Ensuring **phone number has exactly 10 digits**
 - Making sure all fields are filled before submission
- Display appropriate **error messages** near invalid fields.
- Save all three files (index.html, style.css, script.js) in the current directory and print a message like "Event Registration Form Created Successfully."

#CODE:

```

1 <!doctype html>
2 <html lang="en">
3 <head>
4 <meta charset="utf-8" />
5 <meta name="viewport" content="width=device-width,initial-scale=1" />
6 <title>SR University Tech Fest Registration</title>
7 <style>
8 :root{
9     --bg: #f4f7fb;
10     /* Professional, muted palette */
11     --bg: #eef3f8; /* soft neutral page background */
12     --card: linear-gradient(180deg,#ffffff 0%, #f7fbfb 100%); /* crisp card with subtle cool tint */
13     --card-border: #e6eef8;
14     --panel: #ffffff;
15     --glass: rgba(12,18,33,0.03);
16     --text-strong: #0b1220;
17     --muted-soft: #6b7280;
18     --accent: #3b82f6;
19     --danger: #ef4444;
20     --muted: #6b7280;
21     --success: #10b981;
22     --shadow: 0 6px 20px rgba(18,23,38,0.08);
23     --radius:12px;
24     font-family: Inter, ui-sans-serif, system-ui, -apple-system, "Segoe UI", Roboto, "Helvetica Neue", Arial;
25 }
26
27 html,body{
28     height:100%;

```

```

generated_sites > ai assg-14.html > ...
2 <html lang="en">
220 <body>
294 <script>
296 (function(){
401 form.addEventListener('submit', (e) => {
403     const state = validateAll();
404     if(!state.allValid) return;
405     submitBtn.disabled = true;
406     submitBtn.textContent = 'Submitting...';
407
408     // simulate network request
409     setTimeout(() => {
410         submitBtn.textContent = 'Register';
411         form.reset();
412         // clear classes
413         [nameEl, emailEl, phoneEl, deptEl, eventsGroup].forEach(el=>{
414             el.classList.remove('valid', 'invalid');
415         });
416         eventsGroup.querySelectorAll('.checkbox').forEach(l=>l.classList.remove('checked'));
417         submitBtn.disabled = true;
418         formMsg.textContent = 'Registration successful! Check your email for confirmation.';
419         formMsg.classList.add('show');
420         setTimeout(()=> formMsg.classList.remove('show'), 6000);
421         updateSummary();
422     }, 900);
423 });
424
425 // initial summary/state

```

#OUTPUT:

SR University Tech Fest - Student Registration

Please fill in your details. Fields validate in real-time. You must provide a valid email, a 10-digit phone number, choose your department, and select at least one event.

Name

Department

-- Select Department --

Enter a valid name (letters and spaces only).

Please select your department.

Email

Phone

Enter a valid email address.

Phone must be 10 digits.

Event Selection

☐ Hackathon

☐ Workshop

☐ Seminar

☐ Robotics

☐ Coding Contest

Select at least one event.

You can pick multiple events.

Register

Registration Summary

Name: N/A

Phone: N/A

Department: N/A

Events: N/A

Tip: Ensure email is correct and phone is 10 digits. The form will only enable Submit once all validations pass.

#TASK4:

A startup wants a basic e-commerce product page to display products with prices and an “Add to Cart” button.

- Use Copilot to generate a grid-based product catalog in HTML/CSS.
- Implement a JavaScript “Add to Cart” functionality with Copilot’s guidance.
- Modify Copilot’s suggestions to include a cart counter at the top-right corner of the page.

#PROMPT:

- A **grid-based product catalog** displaying multiple products with their **names, images, prices, and “Add to Cart” buttons**.
- A **cart counter icon** at the top-right corner that updates dynamically whenever a user clicks “Add to Cart.”
- Use **responsive CSS (Flexbox or Grid)** to make the layout look good on both desktop and mobile screens.
- Use **JavaScript** to implement “Add to Cart” functionality and update the cart counter in real time.
- Style the page with **modern, attractive colors, rounded product cards, and hover effects** on buttons.
- The project should include three files — index.html, style.css, and script.js.

#CODE:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6   <title>E-Commerce Store</title>
7   <link rel="stylesheet" href="style.css" />
8 </head>
9 <body>
10   <!-- NAVBAR -->
11   <nav class="navbar">
12     <div class="logo">E-Commerce</div>
13     <div class="cart">
14       <span id="cart-count">0</span>
15     </div>
16   </nav>
17
18   <!-- PRODUCT GRID -->
19   <section class="products">
20     <div class="product-card">
21       
22       <div>Wireless Headphones</div>
23       <div>$12.99</div>
24       <button class="add-to-cart">Add to Cart</button>
25     </div>
26
27     <div class="product-card">
28       
```

```

<div class="product-card">
  <h3>Wireless Headphones</h3>
  <p>₹2,999</p>
  <button class="add-to-cart">Add to Cart</button>
</div>

<div class="product-card">
  
  <h3>Smart Watch</h3>
  <p>₹4,499</p>
  <button class="add-to-cart">Add to Cart</button>
</div>

<div class="product-card">
  
  <h3>Gaming Laptop</h3>
  <p>₹79,999</p>
  <button class="add-to-cart">Add to Cart</button>
</div>

<div class="product-card">
  
  <h3>Digital Camera</h3>
  <p>₹22,999</p>

```

#OUTPUT:



Wireless Headphones

₹2,999

Add to Cart



Smart Watch

₹4,499

Add to Cart



Gaming Laptop

₹79,999

Add to Cart



Digital Camera

₹22,999

Add to Cart

